Systems Analysis and Control

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Lecture 12: Root Locus

Overview

In this Lecture, you will learn:

Definition of Root Locus

- Closing the Loop
- effect of gain on Pole Locations

The Effect of Changes in Gain

- Numerical Examples
  - Pole Locations

A Review of Complex Numbers

- Polar Form
- Multiplication-Division

The root locus is defined by phase angles

- Sum of angles from zeros
- MINUS sum of angles from poles
- Have to add up to 180°

Selecting Gain Parameters

Basic Problem: How to select the parameters, K, T_I, T_D in PID control

$$\hat{G}(s) = \frac{n_G(s)}{d_G(s)} \qquad \hat{K}(s) = \frac{K(T_D s^2 + s + T_I)}{s} = \frac{n_K(s)}{d_K(s)}?$$

Performance Specs restrict pole locations to geometric regions.

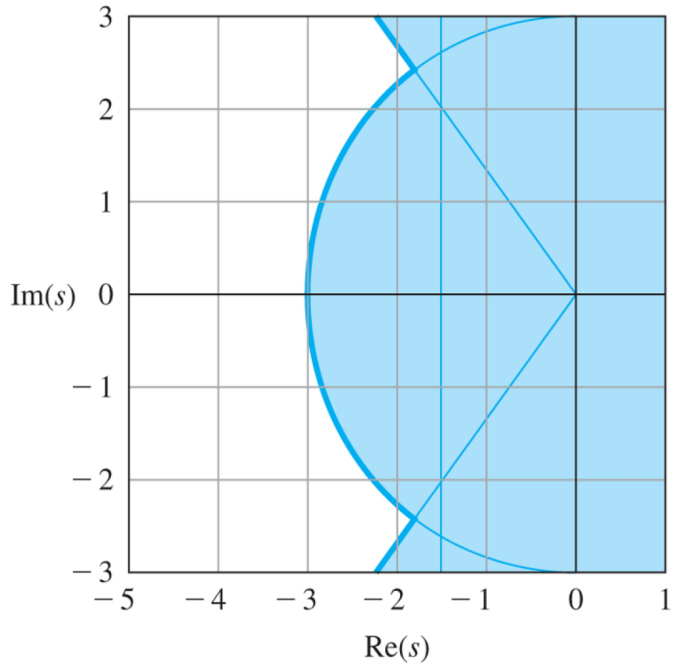
Recall Closed Loop:

$$\frac{GK}{1+GK} = \frac{n_G(s)n_K(s)}{s \cdot d_G(s) + n_G(s)n_K(s)}$$

Question: For given T_I, T_D, how do the poles change with K?

Equivalent Question: What are the roots of

$$s \cdot d_G(s) + K n_G(s)(T_D s^2 + s + T_I)?$$



Question: How to select a value of K so all poles will be in the acceptable region?

Definition of the Root Locus

Given a plant and proportional controller in feedback

$$\hat{G}(s) = \frac{n_G(s)}{d_G(s)} \qquad \hat{K}(s) = k \qquad \frac{k\hat{G}(s)}{1 + k\hat{G}(s)} = \frac{n_G(s)}{d_G(s) + kn_G(s)}$$

Question: What is the set of achievable closed-loop poles?

Definition 1.

Given $n_G(s), d_G(s)$, the **Positive Root Locus** is the set of all roots of

$$d_G(s) + kn_G(s) = 0$$

as k ranges from 0 to ∞

Alternatively:

- The solutions of $\hat{G}(s) = \dfrac{n_G(s)}{d_G(s)} = \dfrac{-1}{k}$ for $k \geq 0$
- Values of s where $\angle G(s) = 180°$ (Woah... Not so fast)

Definition of the Root Locus

Given a plant and proportional controller in feedback

$\hat{G}(s) = \frac{n_G(s)}{d_G(s)}$ $\hat{K}(s) = k$ $\frac{k\hat{G}(s)}{1+k\hat{G}(s)} = \frac{n_G(s)}{d_G(s)+kn_G(s)}$

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Alternatively:

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- Values of s where $\angle G(s) = 180^\circ$ (Woah... Not so fast)

Definition 2.

Given $n_G(s), d_G(s)$, the **Negative Root Locus** is the set of all roots of

$$d_G(s) + kn_G(s) = 0$$

for $-\infty \leq k \leq 0$.

NOTE: The root locus can be used to visualize pole locations for ANY system parameter, c. Just express the denominator of the transfer function as

$$n(s) + cd(s)$$

For example, in PID control, we had $n_K(s) = K(T_D s^2 + s + T_I)$

$$\frac{GK}{1+GK} = \frac{n_G(s)n_K(s)}{s \cdot d_G(s) + n_G(s)n_K(s)}$$

So for fixed K, T_D we could do the root locus for T_I using

$$sd_G(s)+Kn_G(s)(T_D s^2+s+T_I) = \underbrace{(sd_G(s) + Kn_G(s)(T_D s^2 + s))}_{n(s)} + T_I \underbrace{(Kn_G(s))}_{d(s)}$$

What is the Root Locus?

Video Surveillance System.

We can estimate the root locus by finding the roots for many values of k.

Example: Video Surveillance System.
Add a pole at $s = 0$ to eliminate steady-state error.

Open Loop:

$$\hat{G}(s) = \frac{1}{s(s+10)}$$

Closed Loop:

$$\frac{k\hat{G}}{1+k\hat{G}} = \frac{k}{s^2+10s+k}$$

Pole Locations: Exact Formula

$$p_{1,2} = -5 \pm \frac{1}{2}\sqrt{100-4k}$$



(a)

Collecting points on the Root Locus

Video Surveillance System

TABLE 8.1 **Pole location as function of gain for the system of Figure 8.4**

K	Pole 1	Pole 2
0	-10	0
5	-9.47	-0.53
10	-8.87	-1.13
15	-8.16	-1.84
20	-7.24	-2.76
25	-5	-5
30	$-5 + j2.24$	$-5 - j2.24$
35	$-5 + j3.16$	$-5 - j3.16$
40	$-5 + j3.87$	$-5 - j3.87$
45	$-5 + j4.47$	$-5 - j4.47$
50	$-5 + j5$	$-5 - j5$

$$p_{1,2} = -5 \pm \frac{1}{2}\sqrt{100 - 4k}$$

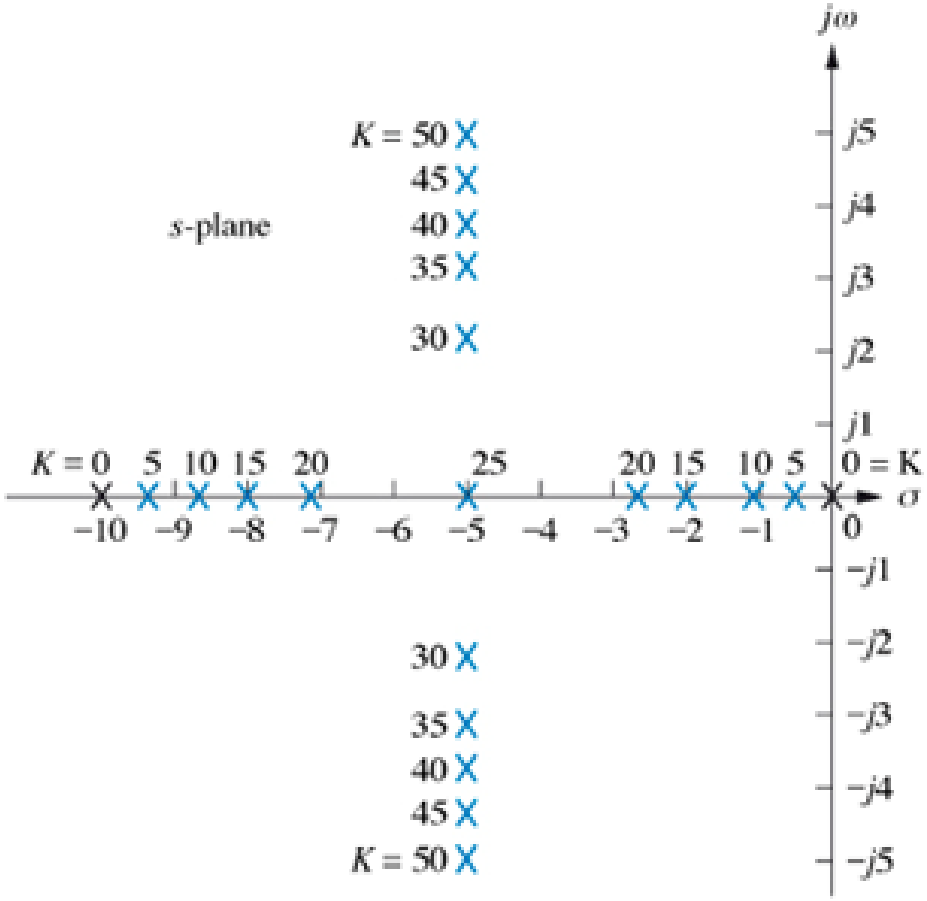
Drawing points on the Root Locus

Video Surveillance System

We visualize the effect of changing k by plotting the poles on the complex plane.

TABLE 8.1 **Pole location as function of gain for the system of Figure 8.4**

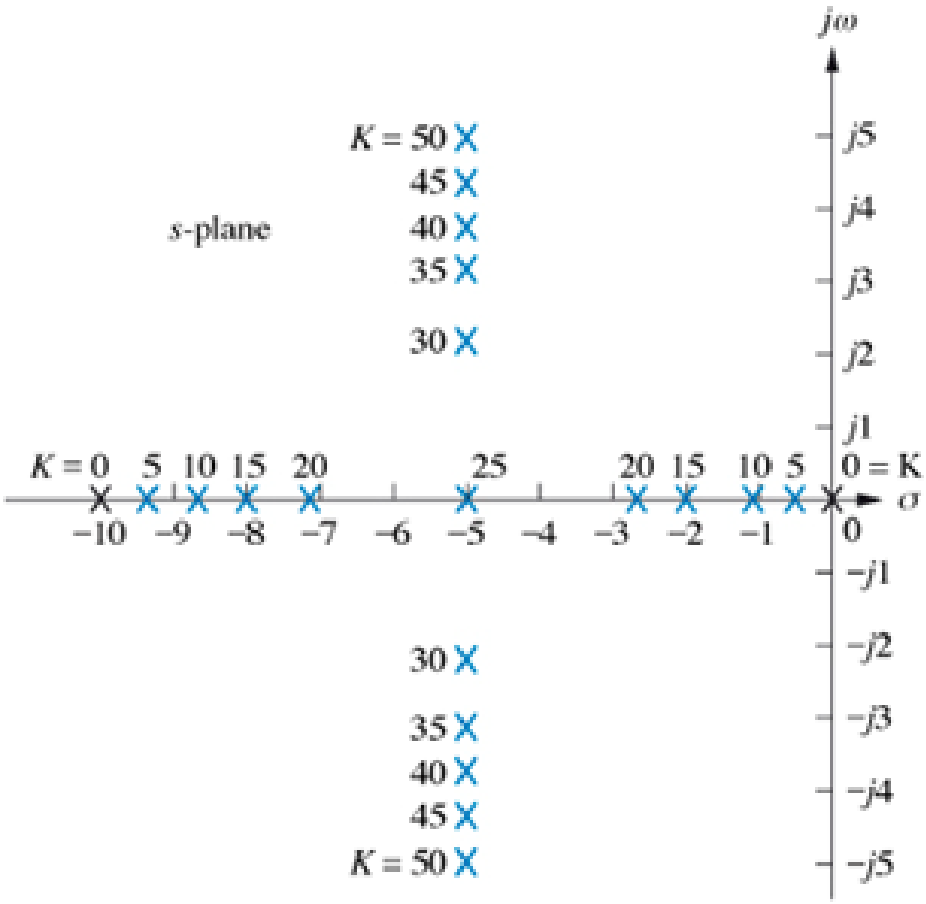
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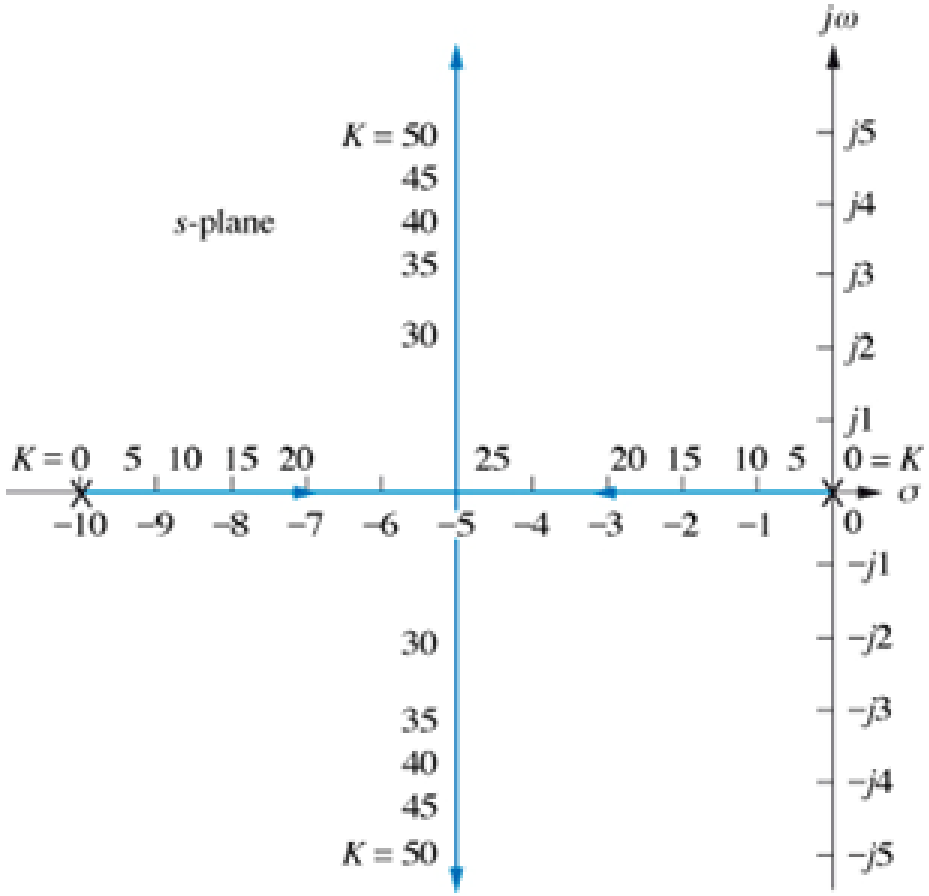


Drawing points on the Root Locus

Connecting the dots to get the root locus

Plotting every possible value of k yields the *root locus*.



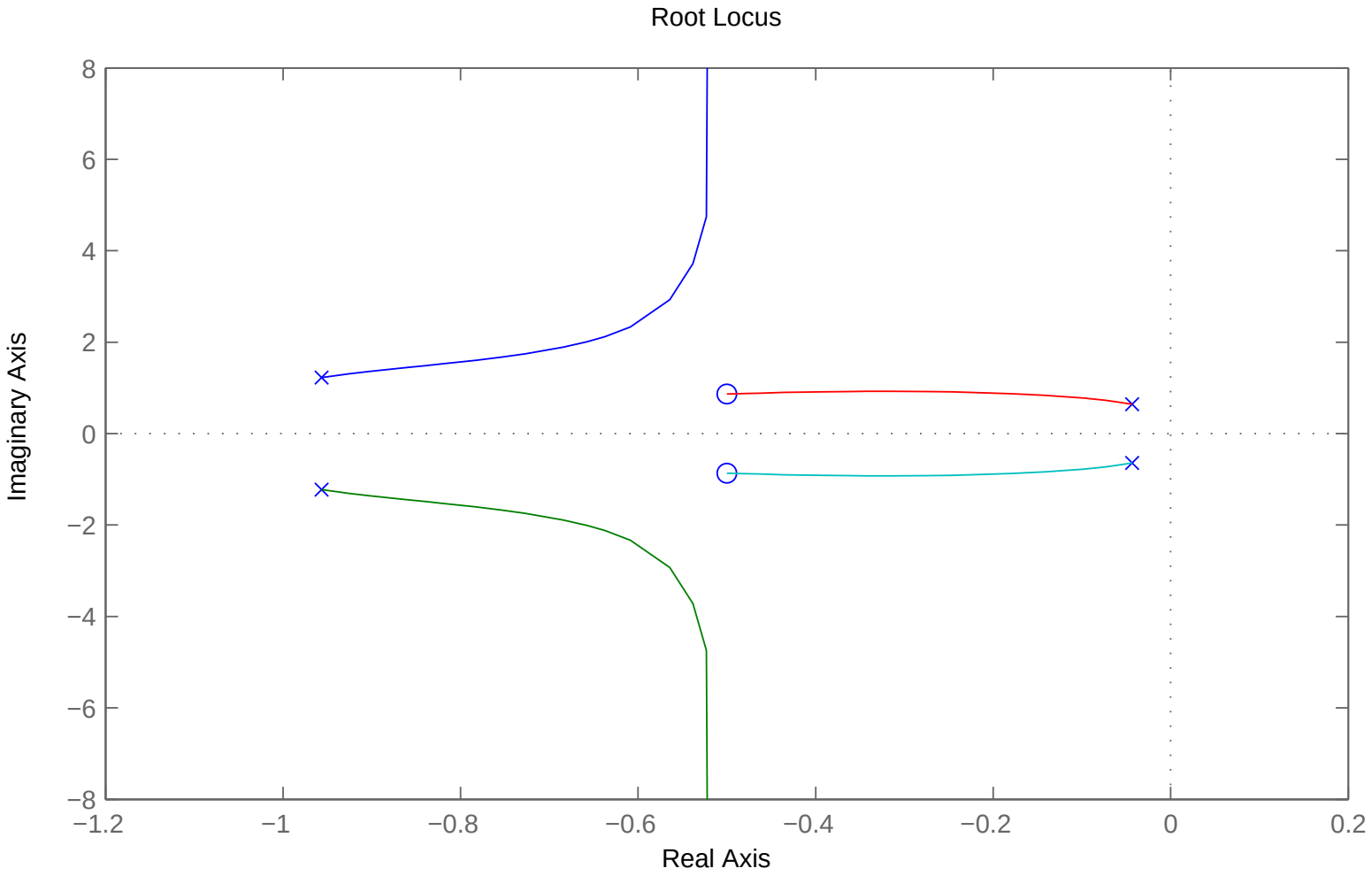


Connect the dots.

Visualizing the Root Locus with MATLAB

Example: Suspension System

Matlab Code: `n_s=[1 1 1]; d_s=[1 2 3 1 1]; rlocus(n_s,d_s)`



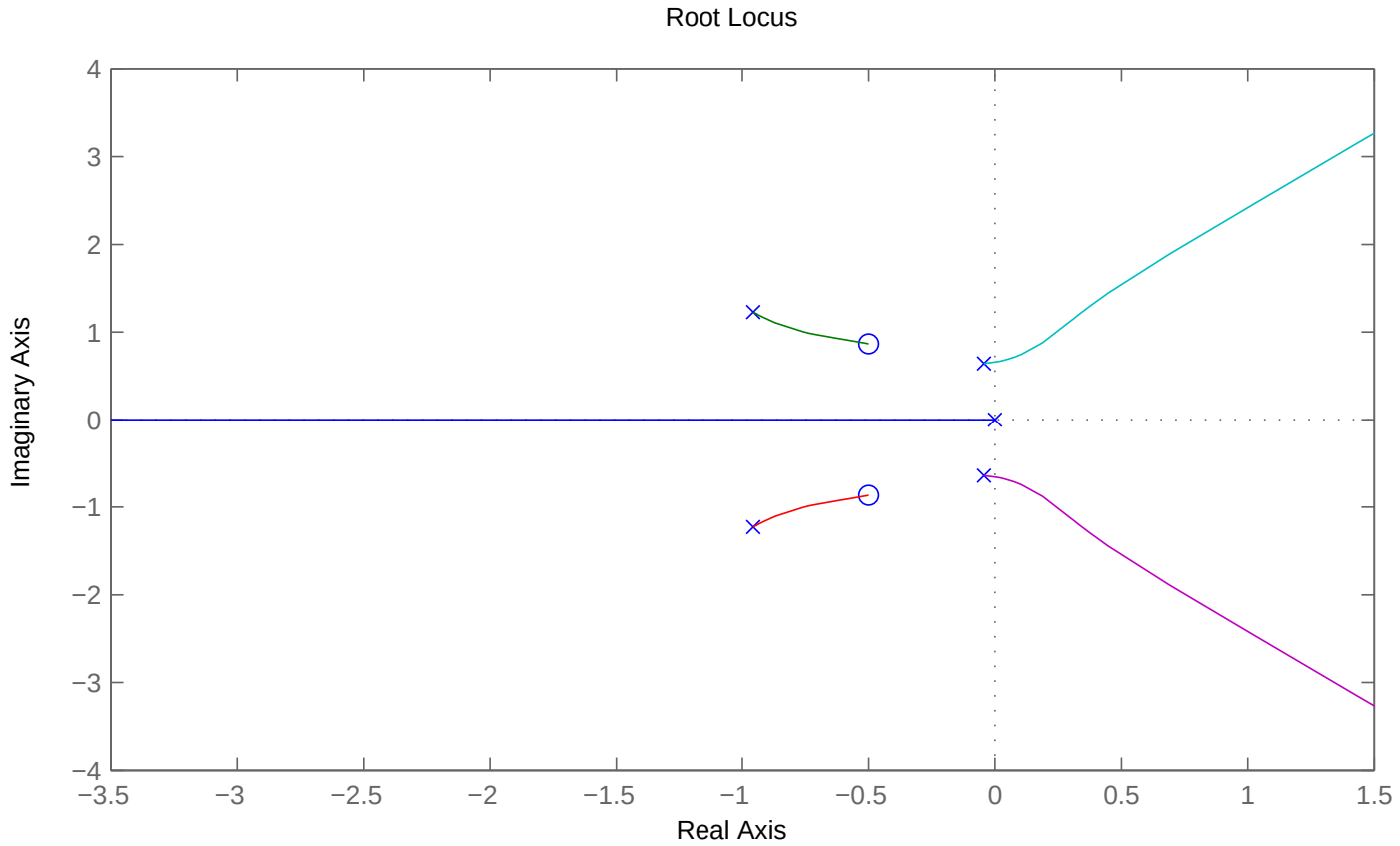
Conclusion: Stable for all $k > 0$.

Visualizing the Root Locus with MATLAB

Example: Suspension System with Integral Feedback

Now, if we add integral feedback: $\hat{K}(s) = k\dfrac{1}{s}$

Matlab Code: `n_s=[1 1 1]; d_s=[1 2 3 1 1 0]; rlocus(n_s,d_s)`



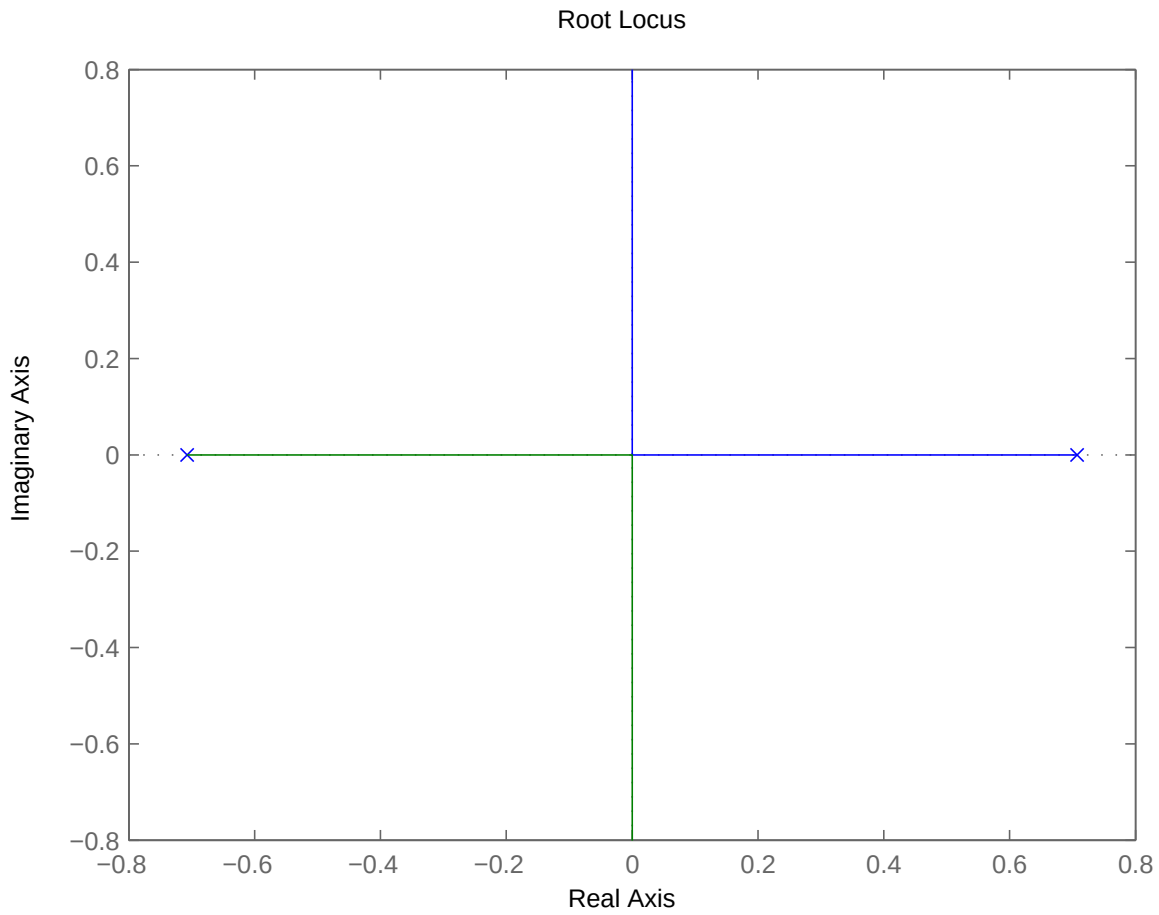
Conclusion: Stable for all $k < .1$.

Visualizing the Root Locus with MATLAB

Example: Inverted Pendulum Model

$$\hat{G}(s) = \frac{1}{s^2 - \frac{1}{2}} \qquad \hat{K}(s) = k$$

Matlab Code: `n_p=[1]; d_p=[1 0 -.5]; rlocus(n_s,d_s)`

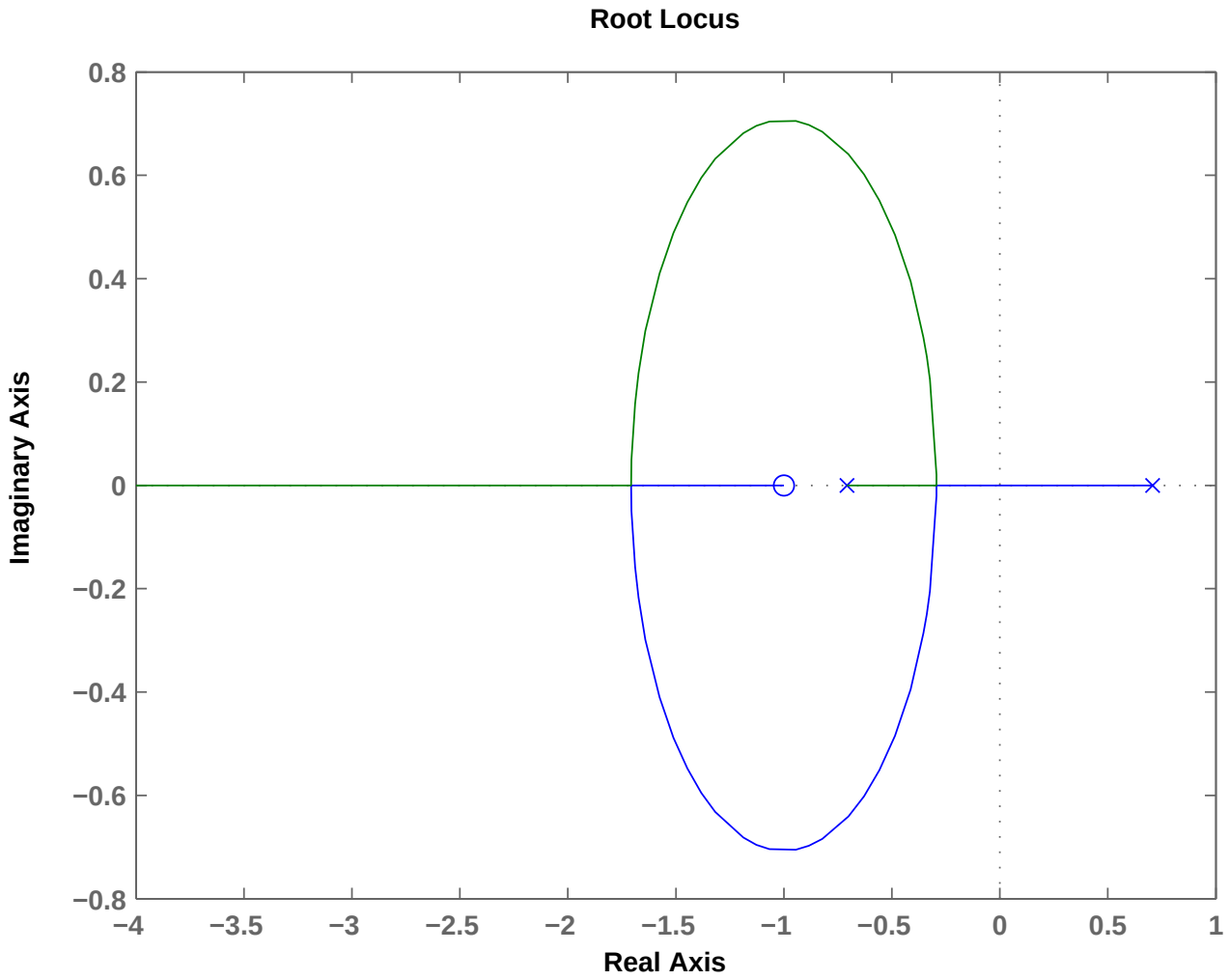


Visualizing the Root Locus with MATLAB

Example: Inverted Pendulum Model with Derivative Feedback

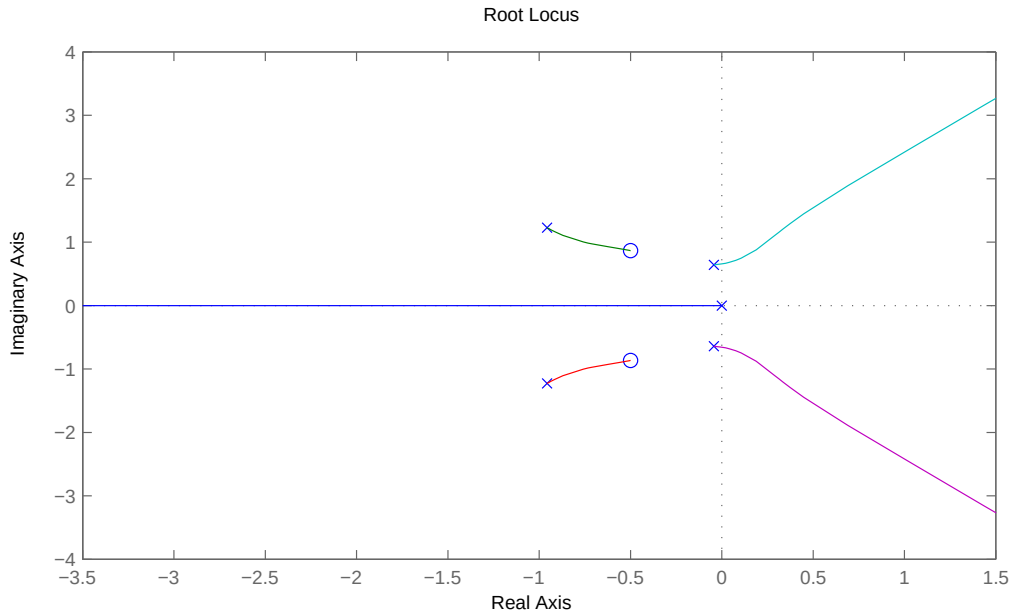
Now an inverted pendulum with some derivative feedback: $\hat{K}(s) = k(1+s)$

Matlab Code: `n_p=[1 1]; d_p=[1 0 -.5]; rlocus(n_s,d_s)`



How to SKETCH the Root Locus without MATLAB?

When Matlab calculates the root locus, it plots every point.



Advantages of knowing how to sketch the root locus:

- Intuition.
  - How open-loop pole AND ZERO locations impact the root locus.
- Suggests how to use controllers to add poles (p_K) and zeros (z_K):

$$\hat{K}(s) = k\frac{s - z_K}{s - p_Z}$$

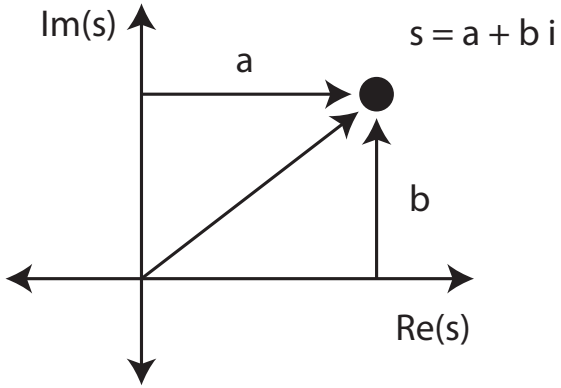
Before sketching the root locus, we briefly review Complex Numbers.

Review of Complex Numbers

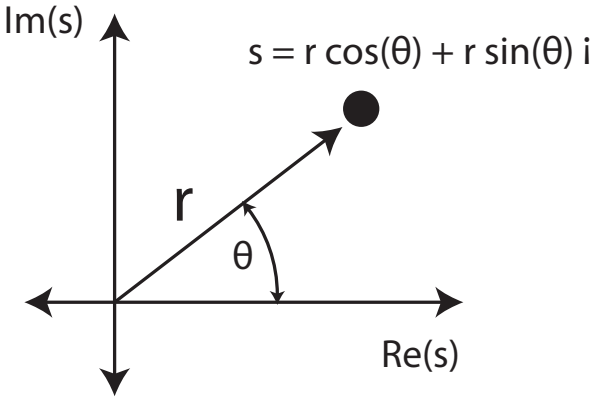
Polar Form of a Complex Number

A Complex Number in Cartesian/Rectilinear Form:

$$s = a + bi$$



So far, we've only used Rectilinear Form.



A complex number can be represented using angle and magnitude:

$$s = r\left(\cos\theta + i\sin\theta\right)$$

Equivalently: **POLAR FORM**

$$s = re^{\theta i}$$

Recall the Euler equation

$$e^{i\theta} = \cos\theta + i\sin\theta$$

Review of Complex Numbers

Magnitude and Phase

Rectilinear

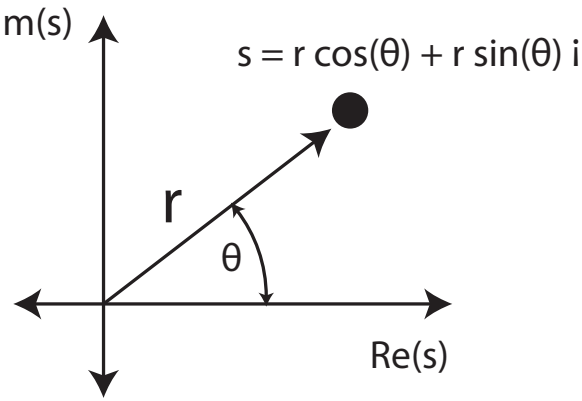
$$s = a + bi$$

Polar

$$s = re^{\theta i}$$

Notation:

- r is called the **Magnitude**
  - Denoted $r = |s|$
- $\theta =$ is called the **Phase**
  - Denoted $\theta = \angle s$



The relationship between Polar and Rectilinear coordinates is obvious

- $\theta = \tan^{-1}\left(\frac{b}{a}\right)$
- $r = \sqrt{a^2 + b^2}$
- $a = r\cos\theta$
- $b = r\sin\theta$

Review of Complex Numbers

Multiplication is Simplified using Polar Form

Consider multiplying two complex numbers, $s_1, s_2 \in \mathbb{C}$ — $s_1 \cdot s_2$

Polar Form of Multiplication:

$$s_1 = r_1 e^{\theta_1 \imath}$$

$$s_2 = r_2 e^{\theta_2 \imath}$$

Multiplication Formula:

$$\begin{aligned} s_1 s_2 &= r_1 e^{\theta_1 \imath} r_2 e^{\theta_2 \imath} = r_1 r_2 e^{\theta_1 \imath} e^{\theta_2 \imath} \\ &= r_1 r_2 e^{(\theta_1 + \theta_2)\imath} \end{aligned}$$

Cartesian Form of Multiplication:

$$s_1 = a_1 + b_1 \imath$$

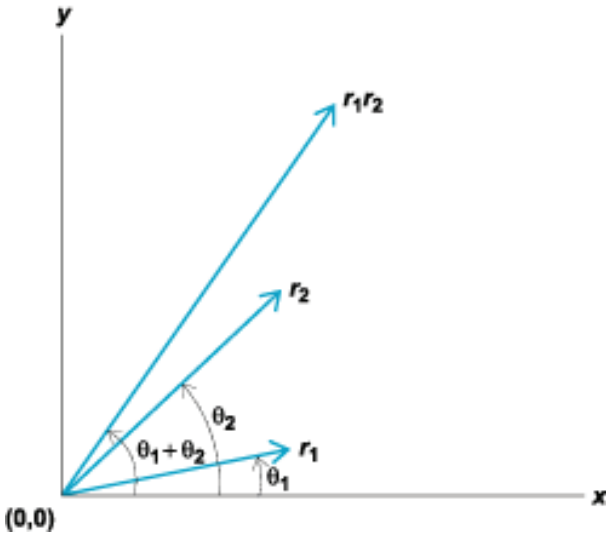
$$s_2 = a_2 + b_2 \imath$$

Multiplication Formula:

$$\begin{aligned} s_1 \cdot s_2 &= (a_1 + b_1 \imath)(a_2 + b_2 \imath) \\ &= (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1)\imath \end{aligned}$$

For multiplication in polar coordinates:

- magnitudes (r_1, r_2) multiply
- phases (θ_1, θ_2) add



Review of Complex Numbers

Division is also simplified using Polar Form

Consider dividing two complex numbers, $s_1, s_2 \in \mathbb{C}$ — $\frac{s_1}{s_2}$

Polar Form of Division:

$$s_1 = r_1 e^{\theta_1 \imath}$$

$$s_2 = r_2 e^{\theta_2 \imath}$$

Division Formula:

$$\frac{s_1}{s_2} = r_1 e^{\theta_1 \imath} r_2^{-1} e^{-\theta_2 \imath} = \\ = \frac{r_1}{r_2} e^{(\theta_1 - \theta_2)\imath}$$

For division in polar coordinates,

- magnitudes (r_1, r_2) divide
- phases (θ_1, θ_2) subtract

Cartesian Form of Division:

$$s_1 = a_1 + b_1 \imath$$

$$s_2 = a_2 + b_2 \imath$$

Division Formula:

$$\frac{s_1}{s_2} = \frac{a_1 a_2 + b_1 b_2}{a_2^2 + b_2^2} + \frac{b_1 a_2 - a_1 b_2}{a_2^2 + b_2^2} \imath$$

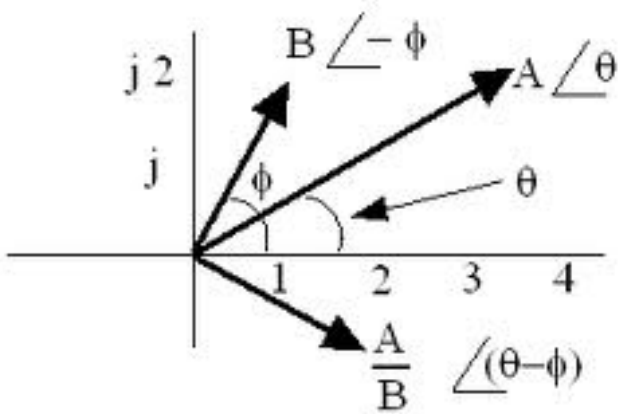


Fig. 17: Division using Polar Form

Interpretation of Root Locus using Polar Coordinates

Root Locus is Points, s, where $\angle G(s) = 180°$

Recall the root locus is the set of s such that

$$n(s) + kd(s) = 0 \qquad \text{OR} \qquad 1 + k\hat{G}(s) = 0 \qquad \text{for SOME} \qquad k \geq 0.$$

In other words,

$$\hat{G}(s) = -\frac{1}{k} \qquad \hat{G}(s) \text{ is a } \textbf{negative real number}$$

In polar coordinates, this is a constraint on phase: $\angle G(s) = \theta = \pi = 180°$

$$\hat{G}(s) = \frac{1}{k} e^{\pi \imath}$$

- Magnitude is $1/k$
- Phase is πrad $= 180°$

Since k can be anything greater than 0:

- Root locus is all points such that

$$\angle \hat{G}(s) = \angle \frac{n(s)}{d(s)} = 180° \pm n \cdot 360°$$

Angles from Zeros and Poles define the Root Locus

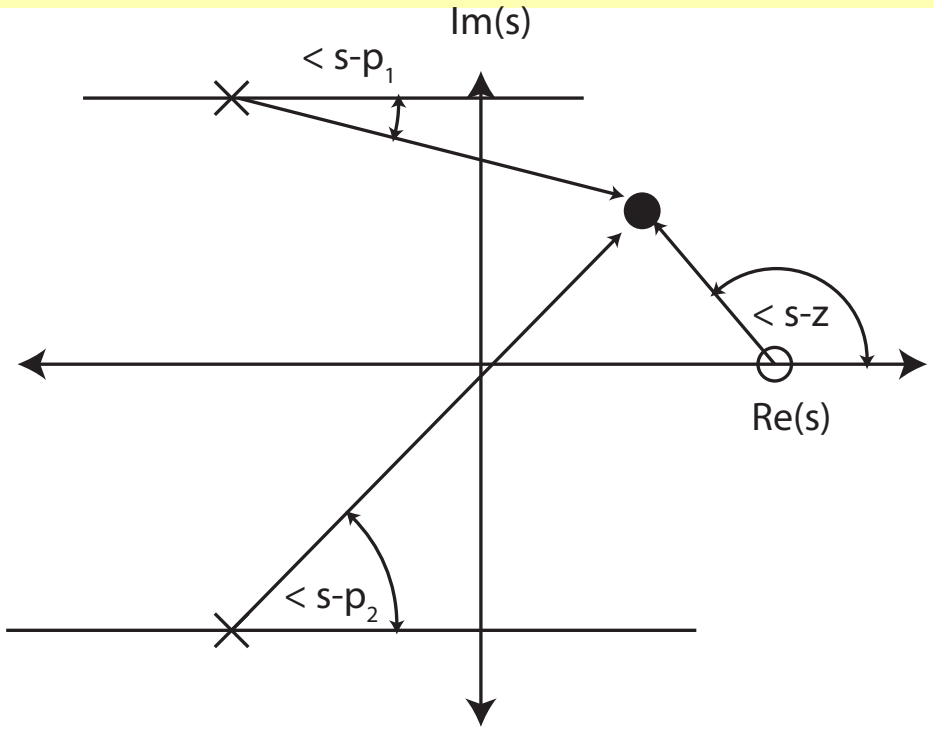
Angles from Zeros MINUS angles from Poles must total 180°

Write the poles and zeros of $\hat{G}(s)$ as

$$\hat{G}(s) = \frac{n(s)}{d(s)} = \frac{(s-z_1)\cdots(s-z_m)}{(s-p_1)\cdots(s-p_n)}$$

Then the phase of $\angle G(s)$ is given by

$$\begin{aligned}\angle\hat{G}(s) &= \angle(s-z_1) + \cdots + \angle(s-z_m) \\ &\quad - \angle(s-p_1) - \cdots - \angle(s-p_n) \\ &= \sum_{i=1}^{m} \angle(s-z_i) - \sum_{i=1}^{n} \angle(s-p_i)\end{aligned}$$



For a point on the root locus: we need

$$\sum_{i=1}^{m} \angle(s-z_i) - \sum_{i=1}^{n} \angle(s-p_i) = 180^\circ \pm n \cdot 360^\circ$$

The angles from zeros **MINUS** the angles from poles have to add up to 180° (plus multiples of $\pm 360^\circ$)

NOTE: Angles are measured Counter-Clockwise (CCW)

Summary

What have we learned today?

Definition of Root Locus

- Closing the Loop
- effect of gain on Pole Locations

The Effect of Changes in Gain

- Numerical Examples
  - Pole Locations

A Review of Complex Numbers

- Polar Form
- Multiplication-Division

The root locus is defined by phase angles

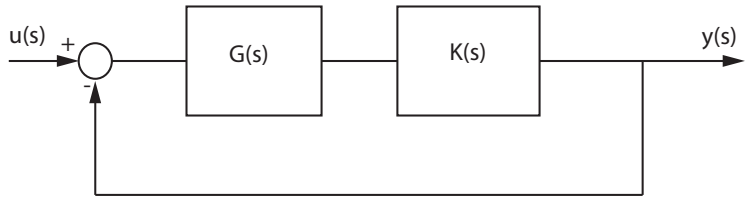
- Sum of angles from zeros
- MINUS sum of angles from poles
- Have to add up to 180°

Next Lecture: Sketching the Root Locus

The Effect of Feedback

Feedback changes the open loop

$$\hat{G}(s) = \frac{n_G(s)}{d_G(s)} \qquad \hat{K}(s) = \frac{n_K(s)}{d_K(s)}$$



to

$$\frac{\hat{G}(s)\hat{K}(s)}{1+\hat{G}(s)\hat{K}(s)} = \frac{n_G(s)n_K(s)}{d_G(s)d_K(s)+n_G(s)n_K(s)}$$

The pole locations are the roots of

$$d_G(s)d_K(s) + n_G(s)n_K(s) = 0$$

Objective: A closed loop denominator.

$$d(s) = (s-p_1)\cdots(s-p_n)$$

Big Question: How to choose $n_K(s)$ and $d_K(s)$ so that

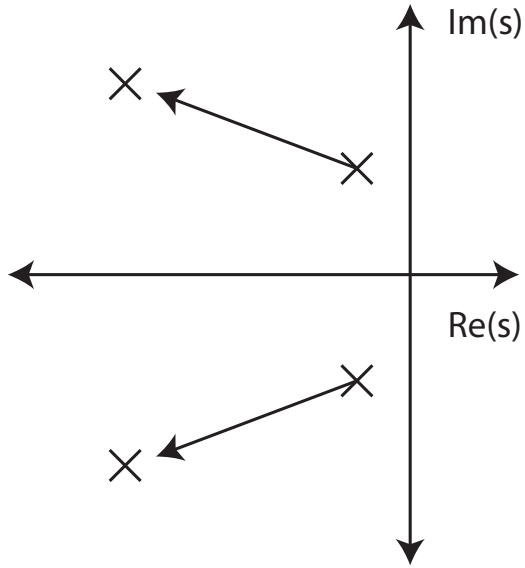
$$d(s) = d_G(s)d_K(s) + n_G(s)n_K(s)$$

The Effect of Feedback

PD Control

For a **Second-Order System**

$$\hat{G}(s) = \frac{1}{s^2 + as + b}$$



with PD feedback

$$\hat{K}(s) = K\left(1 + T_D s\right)$$

We can achieve any denominator

$$d(s) = s^2 + cs + d$$

Question: What happens for more complicated systems:

The Effect of Feedback

Suspension Problem

Open Loop:

$$\frac{s^2+s+1}{s^4+2s^3+3s^2+s+1}$$

Closed Loop:

$$\frac{K\left(1+T_D s\right)\left(s^2+s+1\right)}{s^4+2s^3+3s^2+s+1+K\left(1+T_D s\right)\left(s^2+s+1\right)}$$

$$=\frac{K\left(T_D s^3+(1+T_D)s^2+(1+T_D)s+1\right)}{s^4+(2+KT_D)s^3+(3+K+KT_D)s^2+(1+K+KT_D)s+1+K}$$

Given a desired denominator

$$d(s)=s^4+as^3+bs^2+cs+d$$

Which gives 4 equations and 2 unknowns

$$a=2+KT_D \qquad b=3+K+KT_D$$

$$c=1+K+KT_D \qquad d=1+K$$

There is No Solution!!!.

The Effect of Feedback

Proportional Feedback

More fundamentally, how does changing $n_K(s)$ and $d_K(s)$ change the roots of

$$d_G(s)d_K(s) + n_G(s)n_K(s)?$$

The answer is complicated

- Must account for the effect of each term in n_K and d_K

So simplify, lets consider a controller with only a single free parameter.

$$\hat{K}(s) = k$$

Other options include:

- **PD Control**: $\hat{K}(s) = 1 + T_D s$
- **PI Control**: $\hat{K}(s) = 1 + \frac{1}{T_I s}$

Question: How do the roots of

$$d_G(s) + kn_G(s)?$$

change with k?